

REPORT: ASSIGNMENT 2

C H SUDHANVA
250295

Problem: What are we building and why is it useful?

An automated building detection system using satellite imagery and the YOLOv8 object detection model. Given a satellite image, the system identifies buildings, draws bounding boxes around them, and estimates the total number of buildings present. Accurate building detection can help identify developed areas, estimate parcel utilization, property mapping etc which have direct applications to our project.

DATA

where it came from?

<https://www.kaggle.com/datasets/pallav23/icg-spacenet-sandbox-yolo>

what does it contain?

The dataset used in this project consists of 1,148 satellite image tiles along with corresponding annotation files in YOLO format. Each image is paired with a text file containing bounding box coordinates for all buildings present in the image.

Total images = 1148

Total label = 1148

Example for how the labels look like:

|class|center_x|center_y|width|height|

```
0 0.729077 0.435285 0.233600 0.149215
0 0.715269 0.532808 0.055677 0.031985
0 0.130754 0.210046 0.261508 0.205262
0 0.937254 0.962623 0.120015 0.074754
0 0.854754 0.609085 0.137415 0.083615
0 0.087123 0.959208 0.174246 0.081585
```

in this label there are 6 building, (0 = building class)

where all coordinates are **normalized between 0 and 1 relative to the image dimensions.*

MODEL

YOLOv8 is a real-time object detection model developed by Ultralytics. It predicts object locations and classes directly from images in a single forward pass.

For this project, the **YOLOv8s (small)** variant was selected because it provides a good balance between training speed and detection performance. YOLOv8s contains approximately **11 million parameters**, making it suitable for training on a **Tesla T4 GPU**

TRAINING

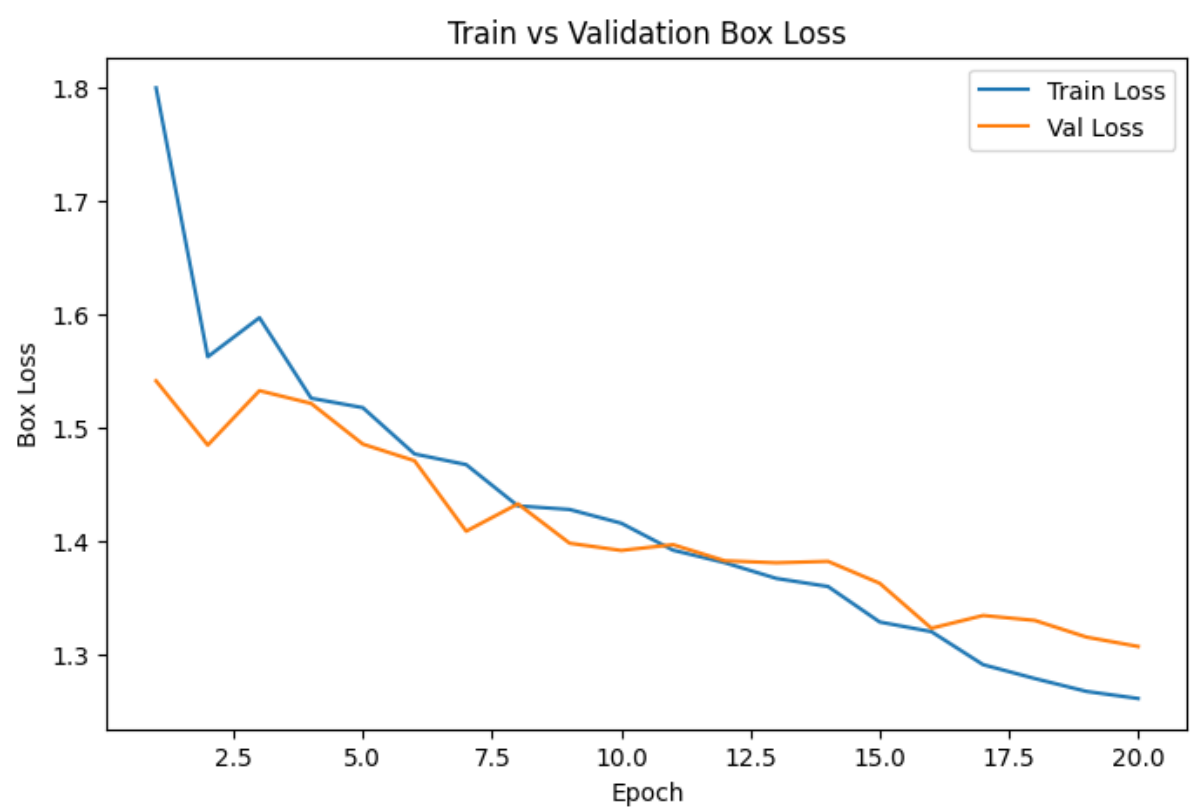
The model was trained using the SpaceNet building detection dataset on a **Tesla T4 GPU** **20 epochs** completed in **0.145 hours** with a **batch size of 16** and **image size of 640*640 pixels**.

RESULTS

METRICS TABLE:

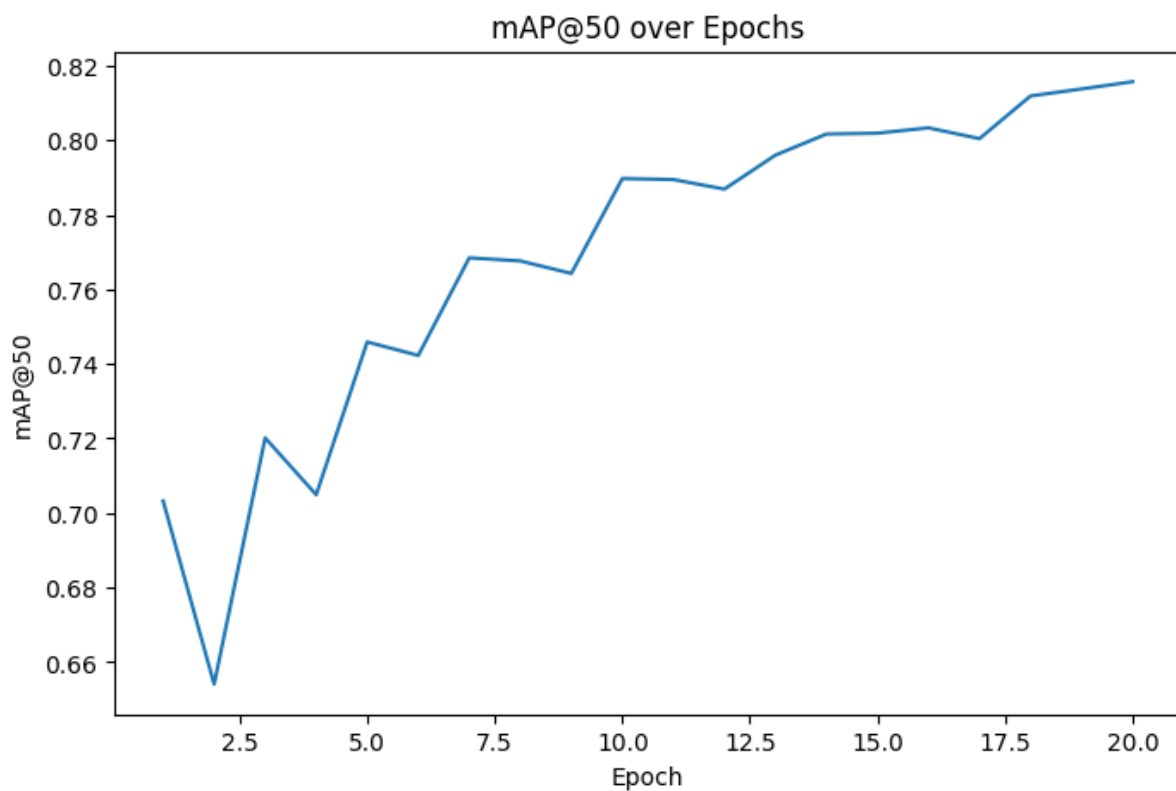
METRIC	Value	What it means
mAP50	0.798	it was able to detect and localize buildings with about 80% overall detection quality when predictions were considered correct at 50% bounding-box overlap
Precision	0.819	82% of predicted buildings were actually buildings
Recall	0.724	72% of all true buildings were detected
F1 Score	0.769	77% being the combined measure of Precision and Recall

TRAIN VS VALIDATION BOX LOSS:



The validation loss closely followed the training loss throughout training, indicating that the model generalized well and did not exhibit significant overfitting.

mAP@50 Curve:



The mAP@50 improved rapidly during the initial epochs and then plateaued, suggesting that the model had largely converged.

SAMPLE PREDICTION IMAGE:

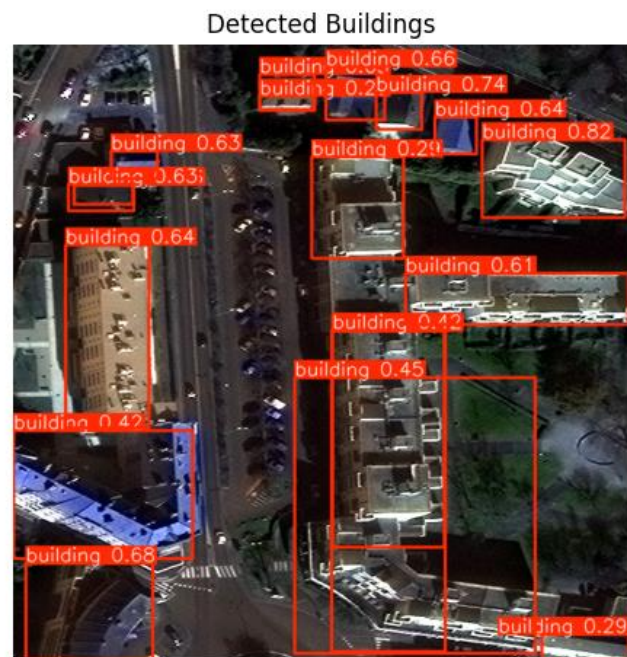


Image: AOI_3_Paris_img1270.png

Buildings detected: 18

Avg confidence: 0.53

Processing time: 36.8 ms

OBSERVATION

What worked well

The model successfully detected most buildings across different urban scenes and achieved strong performance(Map50= 0.798,F1 Score = 0.769) . It learned quickly because we used a pre-trained YOLOv8 model(training time only 0.145 hours) , and its performance improved steadily during training.(did not exhibit significant overfitting, Map50 improved rapidly and also plateaued by 20 epochs)

What the model struggled with

The model sometimes had difficulty detecting very small buildings and buildings that were very close together and sometimes generated overlapping boundaries. It also got confused by some ground patterns or structures that looked similar to buildings(the shape, sometimes shadows, sometimes other structures/vehicles resembling buildings, and low resolution giving shapes, texture to barren lands). Images with large empty areas were sometimes harder to analyze.

One idea to improve it

The model could be improved by training it for more epochs, using a larger YOLO model, or using higher-quality images. More training data and image augmentation techniques could also help improve accuracy.